## 1 Configurations



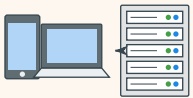
**REF service**
4,096 configs

141 drawn by utilisation of the busiest tier:

- below 85 %: 90
- 85–97 %: 30
- 97 % and over: 20

+ the optimum; 6 configs spanning $U$ for noise

## 2 Two measurements



**(a) emulation**: SimPy, 6,000 sessions (≈3 min)

Erlang-C, M/M/$c$ per tier

**(b) analytic model**: steady state, ground truth



**(c) noise**: 4 sizes, 500 to 4,000 sessions; 20 replications; emulator vs sampler

## 3 Runs



(a)+(b): 141 emulated experiments

(c): 6 configs × 4 sizes × 20 = 480 replications

**621 runs**

## 4 Measured



conversion rate
p95 page time
utility $U$

s.d. of a test result at each size $n$

## 5 Analysis

Correlation and mean difference by stratum

model-error s.d. net of measurement noise

log variance ratio emulator / sampler